Advanced Multivariate Logistic Regression

Predicting Safe Partner Screening Practices

This analysis utilizes a Multivariate Logistic Regression model to identify the *independent* predictors of safe clinical practice (screening one's partner *before* marriage).

By analyzing all variables simultaneously, we calculate the **Adjusted Odds Ratio (aOR)**. This tells us the exact mathematical likelihood of a participant engaging in safe practice for every 1-unit increase in a predictor, *while holding all other demographic variables completely equal*.

Model Summary

- **Dependent Variable:** Safe_Practice (1 = Screened before marriage, 0 = Delayed/Unsafe)
- **Sample Size (N):** 82
- **Pseudo R-Squared:** 0.305

Regression Results: Adjusted Odds Ratios

Predictor Variable	Adjusted Odds Ratio (aOR)	95% CI Lower	95% CI Upper	P-Value	Significant?
Age	**0.837**	0.733	0.957	0.0090	Yes (*)
Is_Married	**1.973**	0.335	11.626	0.4530	No
Education_Level	**1.759**	0.953	3.246	0.0710	No
Knowledge	**1.078**	0.911	1.276	0.3810	No
Attitude	**0.992**	0.626	1.572	0.9730	No

Key Clinical Interpretations

- **Age is the Only Independent Predictor:** When controlling for all other socioeconomic and clinical variables, the only statistically significant predictor of screening one's partner before marriage is **Age** (p = 0.009).
- **The "Younger Generation" Effect:** The Adjusted Odds Ratio (aOR) for Age is **0.837**. Because this is less than 1.0, it means that for every 1-year increase in age, a participant is *less* likely to have screened their partner before marriage. This strongly implies that younger generations are adopting safer screening practices than older generations, likely due to recent public health campaigns.
- **Loss of Significance in KAP:** `Knowledge` (p = 0.381) and `Attitude` (p = 0.973) both lose their statistical significance in this multivariate model.

While we proved they matter in isolated Chi-Square tests, when they are forced to compete against Age and Education in this model, their independent effect is washed out.

- **Note on Sample Size:** The sample size for this specific regression dropped to $N = 82$ because it only includes participants who provided a clear answer for Q33 (when they screened their partner). This smaller sample size contributes to the wider confidence intervals.